

KeyConcept Binomial Theorem

If n is a natural number, then $(a + b)^n =$

$${}_nC_0 a^n b^0 + {}_nC_1 a^{n-1} b^1 + {}_nC_2 a^{n-2} b^2 + \dots + {}_nC_n a^0 b^n = \sum_{k=0}^n \frac{n!}{k!(n-k)!} a^{n-k} b^k.$$

المفهوم الأساسي نظرية ذات الحدين

إذا كان n عددًا طبيعيًا، فإن $(a + b)^n =$

$${}_nC_0 a^n b^0 + {}_nC_1 a^{n-1} b^1 + {}_nC_2 a^{n-2} b^2 + \dots + {}_nC_n a^0 b^n = \sum_{k=0}^n \frac{n!}{k!(n-k)!} a^{n-k} b^k$$

Ex2: Expand $(3x + 2y)^5$. $\rightarrow n=5$

$$= {}_5C_0 (3x)^5 (2y)^0 + {}_5C_1 (3x)^4 (2y)^1 + {}_5C_2 (3x)^3 (2y)^2 + {}_5C_3 (3x)^2 (2y)^3 \\ + {}_5C_4 (3x)^1 (2y)^4 + {}_5C_5 (3x)^0 (2y)^5$$

$$= 243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 \\ + 240xy^4 + 32y^5$$

$$(y - 4z)^4 \quad n=4$$

$$= {}_4C_0 (y)^4 (-4z)^0 + {}_4C_1 (y)^3 (-4z)^1 + {}_4C_2 (y)^2 (-4z)^2 + \\ {}_4C_3 (y)^1 (-4z)^3 + {}_4C_4 (y)^0 (-4z)^4$$

$$= y^4 - 16y^3z + 96y^2z^2 - 256yz^3 + 256z^4$$

Determining specific terms: إيجاد حد معين

$$nC_k (a)^{n-k} \cdot (b)^k$$

Ex3: Find the indicated term of each expression.

23. third term of $(x + 2z)^7$



$$k=3-1=2, \quad n=7, \quad a=x, \quad b=2z$$

$$\begin{aligned} 3^{\text{rd}} \text{ term} &= 7C_2 \cdot (x)^{7-2} \cdot (2z)^2 \\ &= 84x^5z^2 \end{aligned}$$

24. fourth term of $(y - 3x)^6$

$$k=4-1=3, \quad n=6, \quad a=y, \quad b=-3x$$

$$\begin{aligned} 4^{\text{th}} \text{ term} &= 6C_3 \cdot (y)^{6-3} \cdot (-3x)^3 \\ &= -540y^3x^3 \end{aligned}$$

26. sixth term of $(4x + 5y)^6$

$$k=6-1=5, \quad n=6, \quad a=4x, \quad b=5y$$

$$\begin{aligned} 6^{\text{th}} \text{ term} &= 6C_5 \cdot (4x)^{6-5} \cdot (5y)^5 \\ &= 75000xy^5 \end{aligned}$$

